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1) Coin Toss - Maximum Likelihood Estimation (MLE)

- Total coin tosses = 10
- outcome = HHTHHHTHHT
- Number of heads = 7
- number of tails = 3

Formula

$$P(H) = \frac{\text{number of heads}}{\text{Total tosses}}$$

$$P(H) = \frac{7}{10}$$

$$P(H) = 0.7$$

Maximum likelihood Estimation (MLE) is a statistical method used to estimate the most likely value of an unknown probability based on observed data. For a coin toss, the probability of getting head is estimated by divided the number of heads by the total number of tosses.

Based on the observed tosses, the estimated probability of getting heads is 70%.

2) Exam Results - Maximum likelihood Estimation (MLE)

Given

- Total number of students = 50
- Number of students who passed = 42
- Number of students who failed = 8

MLE estimate the probability of an event using the observed number of successes and the total number of observation

Passing the exam is considered a success. The probability of is calculated by dividing the number of student who passed by total number of student.

$$P(\text{Pass}) = \frac{\text{number of student passed}}{\text{Total number of student}}$$

Substituting given values

$$P(\text{Pass}) = \frac{42}{50}$$

$$P(\text{Pass}) = 0.84$$

$$\text{Percentage} = P(\text{Pass}) \times 100$$

$$= 0.84 \times 100$$

$$= 84\%$$

- Estimated probability of passing = 0.84
- percentage of students who passed = 84%

Based on the sample data, the estimated probability that a student passes the exam is 0.84, meaning that 84% of the students passed.

3) Defective Bulbs - Maximum Likelihood Estimation (MLE)

Given

- Total number of bulbs = 100
- Number of defective bulbs = 8
- Number of working bulbs = 92

MLE is used to estimate the probability of an event from the available sample data. Here, a defective bulb is considered the event of interest.

$$P(\text{Defective}) = \frac{\text{number of defective bulbs}}{\text{Total number of bulbs}}$$

$$P(\text{Defective}) = \frac{8}{100}.$$

$$P(\text{Defective}) = 0.08 \times 100 = 8\%.$$

The MLE estimate of the probability of a bulb being defective.

4) Dice Rolls - Maximum Likelihood Estimation (MLE)

- Total number of die rolls = 60
- Number of times six appears = 18
- Number of times other outcomes appear = 42

MLE estimates the probability of an event by using its observed frequency. In this problem.

Formula.

$$P(6) = \frac{\text{number of times six appears}}{\text{Total number of rolls}}$$

$$P(6) = \frac{18}{60}$$

$$P(6) = 0.3$$

$$P(6) = 30\%.$$

The MLE estimate of the probability of getting a six is 0.3 or 30%.

5) Single weight Neuron and Gradient Descent

Given

input (x) = 4

initial weight = 2

Actual weight = 10

learning rate (η) = 0.01

Gradient = 4

Predicted output

$$\hat{y} = w \times x$$

$$\hat{y} = 2 \times 4 = 8$$

Error = Actual output - Predicted output

solution = 10 - 8 = 2

$$w_{new} = w_{old} - \eta \times \text{Gradient}$$

$$w_{new} = 2 - (0.01 \times 4)$$

$$= 2 - 0.04 = 1.96$$

updated weight = 1.96

The predicted output is 8, the error is 2 and the updated weight is 1.96.